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Statement of Purpose

of Tianqi "Gordon" Dai

Machine learning models do not merely predict the world—they reshape it. A credit-scoring algorithm changes who receives loans; a recidivism predictor alters sentencing decisions and, consequently, future crime data. This feedback loop means that deployed ML systems and the societies they touch form a coupled stochastic dynamical system: models shift behavior, and behavior regenerates the data on which future models are trained. My research goal is to build mathematical and computational tools to understand—and ultimately govern—this coupled evolution.

I approach this goal from three directions, as the vertices of a triangle connecting **dynamical systems**, **machine learning**, and **society**. Along one edge, I analyze learning algorithms through dynamical systems. Along a second, I study how choosing among equally accurate models changes social outcomes, and how ML-based simulations can illuminate social dynamics. The third edge—treating deployed models and human responses as one integrated system—is where I want to focus my PhD.

Understanding learning as a dynamical system. To study how models reshape societies, I first studied how models come into being. With Professor Joan Bruna, I extended single-index models [1] to a hierarchical joint-learning setting and reduced high-dimensional SGD to a 3-D coupled ODE tracking three summary statistics. Drawing on *Probability in High Dimension* [2] and work on SGD dynamics [3], [4], I proved that from random initialization SGD escapes a symmetric "mediocre" phase and reaches constant-order overlap with the hidden feature in $O(d \log d)$ steps, while layers synchronize exponentially fast [5]. This project taught me that stochastic learning procedures can often be compressed into tractable dynamical descriptions that still capture the phase transitions that matter.

Leveraging model multiplicity for social impact. A trained model is only one point in a vast landscape: modern ML often yields many near-optimal models that differ in robustness, interpretability, and fairness [6]. With Professors Emily Black and Daniel Neill, I studied this multiplicity via the Rashomon Set (models within ϵ of optimal loss). Inspired by Gibbs sampling for the Ising model, I designed an approximate uniform sampler over classifiers and proved asymptotic results on the set's size and structure [7]. This work received an Honorable Mention for Best Paper at ACM EAAMO 2025 (Top 3/100+). The key message is simple: randomly chosen "good" models can be far from the best available on fairness—selection matters, and can be efficient.

This raised a policy question: if many models are equally accurate, can we choose among them to improve social outcomes? In our setting, selection using sensitive attributes can eliminate demographic disparities within the Rashomon Set, yet anti-discrimination law may forbid such use even when the public demands equal outcomes. Legal scholarship has begun to address this "equal-treatment-versus-equal-outcome" tension [8]. Motivated by this framing, I redesigned the sampler—moving from i.i.d. flips to flips clustered by base rates—and derived variance expressions (and optimizers) for outcome disparities across dataset regimes [9]. The result is a principled way to expand the region of the Rashomon Set that improves disparities without requiring protected attributes at decision time.

Simulating social dynamics via ML-based (LLM) agents. A complementary approach is to study social dynamics directly through simulation. In *Artificial Leviathan* with Professor Misha Sra, I led a team building a survival-game sandbox for LLM agents with Hobbesian drives and bounded memory. Agents farm, trade, ally, and under scarcity sometimes rob; across runs they moved from a "state of nature" toward explicit contracts and authority structures [10], [11]. In our follow-up, *De CivAI*, we introduced direct democracy: agents debate and vote on collective policies (including costly infrastructure against an oncoming flood). These experiments treat LLM societies as stochastic dynamical systems whose macroscopic behavior depends on micro-assumptions, information structure, and institutions.

This work highlighted a prior step: translating social concepts into operational metrics. That motivated me to synthesize a unifying view of a central Responsible AI issue: fairness, privacy, and robustness metrics are often provably inconsistent, both within a concept and across objectives. In my NeurIPS Position Track paper (Top 40/700+) [12], I argued this inconsistency can be constructive—making value pluralism explicit and separating normative trade-offs from statistical ones. This

perspective now shapes how I think about coupled human–model systems: the "dynamics" are not only in parameters and distributions, but also in the evolving metrics that govern deployment.

PhD focus: closing the triangle. The work above has equipped me with tools—dynamical-systems analysis of learning, combinatorial and statistical study of model multiplicity, agent-based simulation of social processes—but each project addressed only one or two edges of the triangle at a time. My PhD goal is to close it: to develop theory and methods for deployed ML and society as a unified coupled stochastic dynamical system.

The performative prediction framework introduced by Perdomo et al. [13] gives a rigorous foundation for exactly this goal: it formalizes how deploying a predictor with parameters θ can induce a data distribution $D(\theta)$, so that the relevant objective is the risk under the distribution created by the model's own deployment. Staying at NYU and working with **Professor Juan Carlos Perdomo**, I would be fortunate to extend this framework to more complex—and more realistic—settings. For example, instead of treating the environment's response as a black-box mapping, I would like to incorporate the epistemic state of individual agents and model how agents act under partial information. This would naturally lead to a distinction between the true response $D(\theta)$ and agents' subjective or misspecified response $\hat{D}(\theta)$, and to studying how $\hat{D}(\theta)$ (and the behaviors it induces) evolve over time. More broadly, while the original performative prediction framework assumes the map $D(\cdot)$ is time-invariant—deploying the same model at different times induces the same distribution —I am interested in regimes where the response itself becomes nonstationary, e.g., $D_t(\theta)$, due to learning, information diffusion, and feedback among agents.

Besides, I'm excited to find the downstream application of this paradigm under the context of AI alignment with **Professor Hua Shen** to model the evolution of bi-directional alignment between human and AI. Last but not least, I'd like to work with **Professor Gautam Kamath** to build better mathematical tool kits by studying problems such as machine unlearning and differential privacy.More broadly, NYU's ecosystem—Courant's applied mathematics, the Center for Data Science, and Tandon's connections to real-world systems—matches my desire to move fluidly between rigorous theory, computational experiments, and questions of social meaning.

In short, I want my PhD to build theory along all three edges of the triangle—and, most centrally, to develop a unified account of deployed ML and society as a coupled stochastic dynamical system. The ultimate aim is not only to characterize how models fit data today, but to reason carefully about how they reshape the worlds in which they are embedded.

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